Structural Heart Disease Detection via Feature Transfer from Echocardiograms to ECG Signals: A Literature Survey

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***Abstract*—Cardiovascular diseases remain the major cause of death worldwide and Structural Heart Disease (SHD) is a central contributor to this cause. Echocardiography is the gold standard for assessing cardiac structure and function, but requires expensive hardware and expert operators. Electrocardiography (ECG), on the contrary, is available and able to test, but has historically limited sensitivity to detect subtle structural abnormalities. Recent advances in deep learning have proved both echocardiogram- and ECG-based models to predict SHD at scale, and emerging work demonstrates that structural information and features encoded in echocardiographic videos can supervise ECG-based models via feature or knowledge transfer. This paper presents a comprehensive literature survey on multimodal cardiac artificial intelligence at the intersection of echocardiography, ECG, and knowledge distillation. We review foundational echo video datasets, state-of-the-art ECG-based SHD prediction models, and latest teacher–student frameworks, including privacy-preserving and data-free variants. We then synthesize open challenges related to uncertainty-aware distillation, cross-modal feature alignment, and position the proposed research—multimodal SHD detection via knowledge transfer from echocardiograms to ECG signals—within these gaps.**

***Index Terms*—Structural Heart Disease, Echocardiography, Electrocardiogram, Deep Learning, Knowledge Distillation, Multimodal Learning.**

I. Introduction

Structural Heart Disease (SHD)—which includes valvular disorders, cardiomyopathies, and ventricular dysfunction—drives significant morbidity and mortality in cardiovascular disease. Appropriate and timely treatment is vital for disease prevention. Currently, clinical practice diagnostic methods are streamlined into two complementary modalities: electrocardiography (ECG), widely used for recording and monitoring cardiac electrical activity and is easily accessible in most primary care and resource-limited settings; and, echocardiography, which generates high-definition, operator-dependent real-time images of the heart's anatomy, and is capable of monitoring the heart in motion.

Echocardiography is deemed the gold standard for evaluation of most structural abnormalities. This includes assessment of left ventricular ejection fraction (LVEF); determination of chamber volumes (size), and, valvular regurgitation. That being said, the cost, availability of infrastructure, and requirements for echocardiography trained sonographers and/or cardiologists are limitations. This is the opposite of the ECG. ECGs are inexpensive, fast, and often the very first test ordered (during outpatient consultations, as an emergency diagnostic, and/or in rural settings), but in the case of structural heart disease (SHD), the evidence of the disease in early stages or the subtle forms is often a significant gap in the interpretation of the test.

Both of these fields have been revolutionized by deep learning. In terms of performance, video-based convolutional networks have also reached the level of being near-expert. Large-scale datasets in ECG have made it possible to model predictions of echoes and multi-disease structural abnormalities from 12-lead waveforms directly. These advances make the case for a multimodal approach: can we use knowledge distillation to transfer the large spatiotemporal knowledge from echocardiograms (teacher) to small deployable ECG models (student)?

The goals of this survey are threefold. To:

- Provide a review of the deep learning models for SHD based on echo and ECG
- Explore the cross-modal and multi-modal learning, primarily focusing on teacher–student distillation
- Provide justification for using a multimodal framework for SHD detection based on knowledge transfer from echocardiograms to ECG on methodological and clinical grounds

II. Clinical Background and Modalities

A. Structural Heart Disease and Diagnostic Workflows

Clinically, SHD (such as aortic stenosis, mitral regurgitation, hypertrophic and dilated cardiomyopathy, and impaired ventricular systolic function) presents as signs and symptoms (e.g. dyspnea, chest pain) and physical findings (e.g. murmurs) as well as abnormalities on imaging. To quantify cardiac structure and function, echocardiography is used to assess LVEF, ventricular volumes, wall motion, valvular morphology, and hemodynamics, forming the basis for many therapeutic decisions.

III. Echocardiography-Based Deep Learning for Structural Heart Disease

A. EchoNet-Dynamic and 3D CNN Models

Let $f_T(\cdot;\theta_T)$ denote a 3D CNN teacher model with parameters θ_T. For an input video V, the model outputs either:

$$\hat{y}_{\text{EF}} = f_T(V;\theta_T) \tag{1}$$

for EF regression, or a probability vector

$$\mathbf{p}_T = f_T(V;\theta_T) \in [0,1]^K \tag{2}$$

for K-class SHD classification. The EchoNet-Dynamic work demonstrates that such models can approximate expert-level performance, providing a robust foundation for supervision of downstream modalities.

B. Lightweight Echo Models and Efficiency

Subsequent work has explored more efficient video architectures. EFNet, proposed by Zhang et al., introduces a lightweight model that reduces parameter count and computational cost while preserving accuracy on the EchoNet-Dynamic benchmark. Using optimized 3D convolution blocks and temporal downsampling strategies, EFNet achieves an MAE around 3.7 and $R^2 \approx 0.82$ for LVEF prediction, showing that high-quality echo teachers need not be prohibitively large. These models are particularly relevant for knowledge distillation: a strong but manageable teacher facilitates training of multiple students (e.g., ECG models) and deployment in resource-constrained environments.

C. Datasets for Echocardiographic AI

Table I summarizes representative echocardiographic datasets and their primary tasks.

TABLE I
Representative Echocardiography Datasets and AI Tasks

Dataset	Modality	Primary Task	Scale
EchoNet-Dynamic	Apical 4-ch video	EF regression, LV segmentation	∼10k studies
Institutional Cohorts	Multi-view echo	SHD classification	10k–100k studies
EFNet Cohorts	Echo videos	Efficient EF estimation	EchoNet-based

IV. ECG-Based Artificial Intelligence for Structural Heart Disease

Structural remodeling of the heart—characterized by chamber enlargement, ventricular wall thickening, and myocardial fibrosis—necessarily alters electrical conduction pathways and produces subtle but systematic changes in depolarization and repolarization patterns. A key line of research maps ECG waveforms directly to echo-derived labels.

A. ECG-to-Echo Label Prediction

A seminal work in this domain is rECHOmmend. The authors constructed a dataset comprising approximately 2 million ECG recordings, each paired with contemporaneous or nearly-contemporaneous echocardiographic assessments. They trained a convolutional neural network $f_S(\cdot;\theta_S)$ that ingests raw 12-lead ECG data and predicts a composite endpoint of structural heart disease detectable by echocardiography.

The student ECG model maps an ECG signal segment E to a probability distribution:

$$\mathbf{p}_S = f_S(E;\theta_S) \in [0,1]^K \tag{3}$$

where K represents the number of disease classes or endpoints. Training employed standard cross-entropy loss:

$$\mathcal{L}_{\text{CE}} = -\sum_{k=1}^{K} y_k \log p_{S,k} \tag{4}$$

where y_k is the one-hot encoded ground-truth label derived from echocardiographic findings. The rECHOmmend model achieved AUROC of 0.91 for composite SHD detection, substantially exceeding the diagnostic accuracy of expert cardiologists reading the same ECGs.

B. Expansion to Multiple Diseases and Population-Level Screening

Following this paradigm, Poterucha et al. (EchoNext) analyzed over 1 million pairs of ECGs and echos, and trained one of the first deep networks for the detection of multi-disease SHD. Their model yielded an overall AUROC of nearly 0.85 across several structural endpoints and outperformed most clinical decision rules. Importantly, the authors thus reported outperforming the cardiologists for some tasks that greatly illustrate the wealth of structural information contained in ECG signals.

Fujiki et al. also trained an ECG-based deep learning model to predict a wide range of echocardiographic abnormalities and for certain of the endpoints, reported AUCs close to 0.89. But they indicated that external validation on independent cohorts showed marked deterioration in performance. This emphasizes the difficulty of generalization and domain shift.

C. ECG to Represent Mechanical Parameters

Rather than having only binary indicators for SHD, some studies try to establish whether more delicate mechanical indicators, such as global longitudinal strain (GLS), may also be predicted from the ECG. Choi et al. developed a deep learning model to estimate strain from 12-lead ECGs and presented that there exist non-negligible correlations between predicted strain and echo-measured strain; these findings are in line with the hypothesis that ECG contains the signals of myocardial mechanics.

V. Multimodal Learning and Knowledge Distillation

A. Teacher–Student Knowledge Distillation Framework

Knowledge distillation is a technique in which a high-capacity teacher model transfers to a lower-capacity student model. The classical formulation of the teacher produces a soft probability distribution q over classes using a temperature-scaled softmax. The student produces its own soft distribution p, and the loss from the distillation is represented with the Kullback-Leibler (KL) divergence as follows:

$$\mathcal{L}_{KD} = T^2 \sum_{k=1}^{K} q_k \log \frac{q_k}{p_k} \tag{5}$$

The training objective consists of the classic cross-entropy $\mathcal{L}_{CE}$ and the distillation term:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{KD} + \mathcal{L}_{CE} \tag{6}$$

with α capturing the relative impact of teacher supervision. In the setting of SHD, for instance, the teacher can be a 3D CNN that was trained on echocardiogram videos, while the student might be a 1D ConvNet or Conv1D+Transformer model trained using the ECG signals. The goal of this cross-modal KD is the process of transferring the knowledge that has been learned from the echo to the ECG representations.

B. Data-Free and Privacy-Preserving Distillation

Clinical data are very sensitive and are often stored separately at different institutions. Chen et al. have put forward the framework of EchoDFKD, which enables KD without data. And, at the same time, the echo segmentation models are able to be learned without having direct access to the original patient data. The teacher is simulated by generating synthetic inputs, and optimization of the student segmentation network is done using the gradients. While EchoDFKD relies on the intra-modality echo-to-echo distillation, which is a proof that KD can effectively be done in the strictest of privacy situations.

Yang et al. proposed a federated model for heart disease prediction using KD. Each local institution has its own private ECG and clinical data, which they use to train the local models. The knowledge will then be collected by a central server through KD rather than raw gradients or sharing data. The federated knowledge distillation (FKD) reduces communication overhead, and also enhances privacy, thus making it a desirable option for multi-centre SHD model development.

C. Uncertainty-Aware Distillation

One of the primary shortcomings of KD techniques is the presumption of equal teacher prediction consistency. This is very rare in medical imaging and the analysis of signals: bad image quality, noise, and out-of-distribution samples can all degrade the teacher's performance. So far, the methodology of explicit uncertainty utilizing echo-to-ECG distillation weighting has not been adequately researched; therefore, it represents a key domain of innovation on SHD detection through multiple modalities.

D. Conceptual Multimodal KD Framework

Theoretical multimodal SHD detection is shown in Figure 1 which shows a high-level architecture using the transfer of knowledge from echocardiograms to ECG signals.

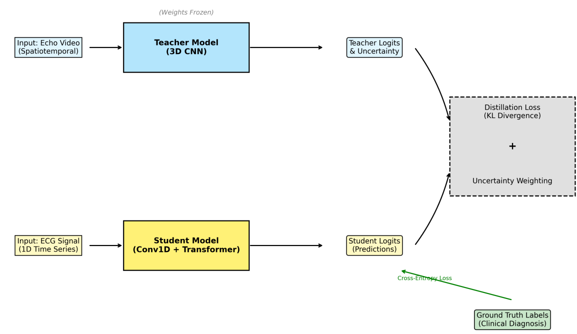

Fig. 1. Schematic teacher-student paradigm: a 3D CNN teacher extracts SHD-relevant embeddings from echocardiogram videos and produces soft labels and uncertainty estimates, while a student model of Conv1D+Transformer takes in the ECG signals and is trained using a mix of cross-entropy loss and uncertainty-weighted distillation loss.

In this model:

- Preprocessing: The teacher processes the video input V and produces logits z_T and uncertainty $u(V)$.
- The student takes aligned ECG sections E and produces logits z_S.
- Distillation may be done either at logit level (soft targets) or even at the feature level (matching intermediate representations).

VI. Synthesis of Research Gaps

A review of the literature also reveals a number of lacunae that justify the proposed research topic.

A. Limited Cross-Modal Feature-Level Distillation

Most ECG-based models for the detection of SHD and guided by echocardiography are based on echo-derived labels rather than echo-derived features. Studies like rECHOmmend and EchoNext consider echo as a provider of global target variables, for instance: reduced EF, valve disease. These studies do not distill spatiotemporal echo features into ECG embeddings. There is an increasing demand for the type of architectures that can match high-dimensional video embeddings with 1D signal embeddings, possibly through shared latent spaces or through attention-based cross-modal transformers.

B. Poor Uncertainty Modeling in KD

Consequently, the importance of calibrating uncertainty has been acknowledged in risk prediction, yet few studies explicitly incorporate teacher uncertainty into the KD objective. To date, echo-to-ECG distillation was not systematically evaluated with explicit weighting of uncertainty. In this, students will could not differentiate between reliable and unreliable teacher predictions. The student will not be able to capture the robustness of safety and also minimizes the overfitting case significantly.

C. Domain Generalization and Institutional Variation

Echocardiograms and ECGs are obtained using different equipment manufacturers' devices, settings, and operator protocols across institutions. Models trained at one center often show degraded performance at another. Knowledge transfer approaches must meet this challenge by domain adaptation techniques, or through learning more generalizable intermediate representations robust to acquisition variation.

VII. Conclusion

AI-driven SHD detection studies conclude that both echocardiogram and ECG independently allow deep learning models of high quality. Echo-based 3D CNNs have almost reached the threshold of expert EF estimation and structural assessment, while ECG-based models have been trained on immense paired ECG–echo datasets to achieve impressive AUROC by inferring SHD labels. Multimodal and knowledge distillation frameworks provide a validated path to combining echocardiography's high-fidelity yet resource-intensive nature along with the accessibility of ECG.

Yet, there are still considerable gaps. Feature-level, cross-modal distillation work rarely performs feature-level, cross-modal distillation from videos to signals; teacher uncertainty is seldom incorporated into the distillation process; and deployment constraints receive insufficient emphasis.

The proposed research—multimodal SHD detection via knowledge transfer from echocardiograms to ECG signals—is well aligned with these open problems. By constructing a 3D CNN echo teacher, a Conv1D+Transformer ECG student, and an uncertainty-aware framework has the potential to produce an effective and reliable tool for large-scale SHD screening in both tertiary and resource-limited environments.

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